Applied Cryptography Assignment 1

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Assignment: 1
Note: I am attempting Questions **1, 2, 3, 5, and 6**.

Notation and small tools used for all question:

- I use alphabet mapping $A = 0, B = 1, \ldots, Z = 25$ (so letters are numbers in $\mathbb{Z}_{26}$). All arithmetic is modulo 26 unless stated.
- When I write $x \equiv y \pmod{26}$, it mean 26 divides $(x - y)$.
- In Hill cipher, convert each block into a column vector using same mapping, e.g., `ACT` $\mapsto [0, 2, 19]^T$.
- A number a has a modular inverse modulo m iff $\gcd(a, m) = 1$. Than $a \cdot a^{-1} \equiv 1 \pmod{m}$.
- For probability, I use $\Pr[\cdot]$. For expectation, I use $\mathbb{E}[\cdot]$.

Question 1 (6-column transposition)

Problem (given):
Plaintext (spaces remove) is writen row-by-row into a grid with 6 columns. If length is not multiple of 6, pad with Z. Then read columns top-to-bottom in secret order $\pi(1), \pi(2), \ldots, \pi(6)$ and concatenate.
Partial info: $\pi(1) = 4$, $\pi(3) = 1$, $\pi(6) = 2$.
Plain text: `ATTACK POSTPONED UNTIL FURTHER ODERS`
Cipher text: `ATUUOZCPNRDZAPNIHRKOTTEZTSDFRZTOELES`

(1) Find the complete permutation π

Step A: Remove spaces and pad.
First remove spaces:

$$P = \texttt{ATTACKPOSTPONEDUNTILFURTHERODERS}.$$

I count the length:

$$L = 32.$$

Since there are 6 columns, need length to be multiple of 6:

$$32 \bmod 6 = 2 \;\Rightarrow\; \text{I need } (6-2) = 4 \text{ Z's}.$$

So the padded plaintext is (length 36):

$$P_{\text{pad}} = \texttt{ATTACKPOSTPONEDUNTILFURTHERODERSZZZZ}.$$

Now:

$$36/6 = \text{So now with6 rows and 6 columns.}$$

Step B: Write into a 6×6 grid row-by-row.

Row	Col 1	Col 2	Col 3	Col 4	Col 5	Col 6
1	A	T	T	A	C	K
2	P	O	S	T	P	O
3	N	E	D	U	N	T
4	I	L	F	U	R	T
5	H	E	R	O	D	E
6	R	S	Z	Z	Z	Z

Step C: Reading each of the columns top-to-bottom.

$$\begin{aligned}
&\text{Column 1: } A\,P\,N\,I\,H\,R = \texttt{APNIHR}\\
&\text{Column 2: } T\,O\,E\,L\,E\,S = \texttt{TOELES}\\
&\text{Column 3: } T\,S\,D\,F\,R\,Z = \texttt{TSDFRZ}\\
&\text{Column 4: } A\,T\,U\,U\,O\,Z = \texttt{ATUUOZ}\\
&\text{Column 5: } C\,P\,N\,R\,D\,Z = \texttt{CPNRDZ}\\
&\text{Column 6: } K\,O\,T\,T\,E\,Z = \texttt{KOTTEZ}
\end{aligned}$$

Step D: Split ciphertext into 6 equal blocks and match.
The Ciphertext is:

$$C = \texttt{ATUUOZCPNRDZAPNIHRKOTTEZTSDFRZTOELES}.$$

Length is of 36, so each of the blocks has 6 letters in it. I split them:

$$B_1 = \texttt{ATUUOZ},\ B_2 = \texttt{CPNRDZ},\ B_3 = \texttt{APNIHR},\ B_4 = \texttt{KOTTEZ},\ B_5 = \texttt{TSDFRZ},\ B_6 = \texttt{TOELES}.$$

Now match each block to column strings:

- `ATUUOZ` is Column 4 $\Rightarrow \pi(1) = 4$ (matches given).
- `APNIHR` is Column 1 $\Rightarrow \pi(3) = 1$ (matches given).
- `TOELES` is Column 2 $\Rightarrow \pi(6) = 2$ (matches given).
- `CPNRDZ` is Column 5.
- `KOTTEZ` is Column 6.
- `TSDFRZ` is Column 3.

So the reading order becomes:

$$\pi(1) = 4,\ \pi(2) = 5,\ \pi(3) = 1,\ \pi(4) = 6,\ \pi(5) = 3,\ \pi(6) = 2.$$

Therefor the full permutation in one-line notation is:

$$\boxed{\pi = (4, 5, 1, 6, 3, 2)}.$$

For Quick Verification: Concatenating columns in order $(4, 5, 1, 6, 3, 2)$ gives us the exact intercept ciphertext.

(2) Encrypt: ABORT MISSION IMMEDIATELY

Step A: Remove spaces and pad.
Spaces removed:

$$P' = \texttt{ABORTMISSIONIMMEDIATELY}.$$

Length:

$$|P'| = 23, \quad 23 \bmod 6 = 5 \Rightarrow \text{I add 1 Z}.$$

So padded plaintext is:

$$P'_{\text{pad}} = \texttt{ABORTMISSIONIMMEDIATELYZ}$$

with length 24. Hence:

$$24/6 = 4 \text{ rows and } 6 \text{ columns}.$$

Step B: Fill the 4×6 grid row-wise.

Row	Col 1	Col 2	Col 3	Col 4	Col 5	Col 6
1	A	B	O	R	T	M
2	I	S	S	I	O	N
3	I	M	M	E	D	I
4	A	T	E	L	Y	Z

Step C: Column strings.

$$\begin{aligned}
&\text{Col 1: } A\,I\,I\,A = \texttt{AIIA}\\
&\text{Col 2: } B\,S\,M\,T = \texttt{BSMT}\\
&\text{Col 3: } O\,S\,M\,E = \texttt{OSME}\\
&\text{Col 4: } R\,I\,E\,L = \texttt{RIEL}\\
&\text{Col 5: } T\,O\,D\,Y = \texttt{TODY}\\
&\text{Col 6: } M\,N\,I\,Z = \texttt{MNIZ}
\end{aligned}$$

Step D: Read columns in order $\pi = (4, 5, 1, 6, 3, 2)$.

$$C' = \texttt{RIEL} \,\|\, \texttt{TODY} \,\|\, \texttt{AIIA} \,\|\, \texttt{MNIZ} \,\|\, \texttt{OSME} \,\|\, \texttt{BSMT}.$$

So final ciphertext is:

$$\boxed{C' = \texttt{RIELTODYAIIAMNIZOSMEBSMT}}.$$

(3) Brute force if attacker only know "6-column transposition"

Attacker must try all permutations of 6 columns:

$$6! = 720.$$

If each try taken by atacker is 1 microsecond $= 10^{-6}$ seconds:

$$T = 720 \times 10^{-6} \text{ s} = 0.72 \text{ ms}.$$

(4) Why no frequency resistance, but still blocks simpel pattern matching? (vs Vigenère)

(a) Frequency analysis still work: Transposition does not change letters, only reorders them. So letter counts stay same.

(b) Pattern matching breaks: Positions change, so full words usually get split. Exact substring search (like `ATTACK`) usually fail.

(c) Vigenère contrast: Vigenère changes letters (so can hide frequency better), but keeps positions, so repetition patterns can still leak info.

Final Answers (Q1):

- $\boxed{\pi = (4, 5, 1, 6, 3, 2)}$
- $\boxed{\texttt{ABORT MISSION IMMEDIATELY} \Rightarrow \texttt{RIELTODYAIIAMNIZOSMEBSMT}}$
- $\boxed{6! = 720 \text{ tries, worst time } = 720\mu s = 0.72 \text{ ms}}$

Question 2 (3×3 Hill cipher key matrix K mod 26)

Problem (given): Alphabet mapping $A = 0, \ldots, Z = 25$. Key matrix K is 3×3. Pairs:

$$\texttt{ACT} \to \texttt{POH}, \qquad \texttt{NOW} \to \texttt{YQF}.$$

All arithmetic is modulo 26.

(0) Hill cipher rule i use

For a 3×3 Hill cipher, I convert each 3-letter block into a column vector of numbers (A=0, B=1, ..., Z=25). Encryption rule is:

$$C \equiv KP \pmod{26},$$

where P is plaintext vector (3×1), C is ciphertext vector (3×1), and K is the 3×3 key matrix.

(1) Convert the given pairs to numbers

Pair 1: `ACT` → `POH`

$$P_1 = \begin{bmatrix} A \\ C \\ T \end{bmatrix} = \begin{bmatrix} 0 \\ 2 \\ 19 \end{bmatrix}, \qquad C_1 = \begin{bmatrix} P \\ O \\ H \end{bmatrix} = \begin{bmatrix} 15 \\ 14 \\ 7 \end{bmatrix}.$$

Pair 2: `NOW` → `YQF`

$$P_2 = \begin{bmatrix} N \\ O \\ W \end{bmatrix} = \begin{bmatrix} 13 \\ 14 \\ 22 \end{bmatrix}, \qquad C_2 = \begin{bmatrix} Y \\ Q \\ F \end{bmatrix} = \begin{bmatrix} 24 \\ 16 \\ 5 \end{bmatrix}.$$

So constraints are:

$$KP_1 \equiv C_1 \pmod{26}, \qquad KP_2 \equiv C_2 \pmod{26}.$$

(2) Solve for K row-by-row (simpel way)

Let one row of K be $[a \;\; b \;\; c]$. Than:

$$[a \;\; b \;\; c]P_1 \equiv d_1 \pmod{26} \Rightarrow 2b + 19c \equiv d_1 \pmod{26},$$

$$[a \;\; b \;\; c]P_2 \equiv d_2 \pmod{26} \Rightarrow 13a + 14b + 22c \equiv d_2 \pmod{26}.$$

Row 1 targets: $(d_1, d_2) = (15, 24)$.
Choose $c = 19$. Than $19c = 361 \equiv 23 \pmod{26}$:

$$2b + 23 \equiv 15 \Rightarrow 2b \equiv 18 \Rightarrow b \equiv 9.$$

Now:

$$13a + 14(9) + 22(19) \equiv 24.$$

Compute $14 \cdot 9 = 126 \equiv 22$, $22 \cdot 19 = 418 \equiv 2$:

$$13a + 22 + 2 \equiv 24 \Rightarrow 13a \equiv 0 \Rightarrow a \text{ even}.$$

Choose $a = 0$. So Row 1 is $[0, 9, 19]$.

Row 2 targets: $(d_1, d_2) = (14, 16)$.

Choose $c = 8$. Than $19c = 152 \equiv 22$:

$$2b + 22 \equiv 14 \Rightarrow 2b \equiv 18 \Rightarrow b \equiv 9.$$

Now:

$$13a + 14(9) + 22(8) \equiv 16.$$

Compute $14 \cdot 9 \equiv 22$, $22 \cdot 8 = 176 \equiv 20$:

$$13a + 22 + 20 \equiv 16 \Rightarrow 13a + 42 \equiv 16.$$

Since $42 \equiv 16$, we get $13a \equiv 0$. Choose $a = 0$. So Row 2 is $[0, 9, 8]$.

Row 3 targets: $(d_1, d_2) = (7, 5)$.
Choose $c = 23$. Then $19c = 437 \equiv 21$:

$$2b + 21 \equiv 7 \Rightarrow 2b \equiv 12 \Rightarrow b \equiv 6.$$

Now:

$$13a + 14(6) + 22(23) \equiv 5.$$

Compute $14 \cdot 6 = 84 \equiv 6$, $22 \cdot 23 = 506 \equiv 12$:

$$13a + 6 + 12 \equiv 5 \Rightarrow 13a + 18 \equiv 5 \Rightarrow 13a \equiv 13.$$

Choose $a = 1$. So Row 3 is $[1, 6, 23]$.

(3) Final key matrix K

$$\boxed{K = \begin{bmatrix} 0 & 9 & 19 \\ 0 & 9 & 8 \\ 1 & 6 & 23 \end{bmatrix} \pmod{26}}$$

(4) Quick verification

Check `ACT` $\to$ `POH`:

$$KP_1 = \begin{bmatrix} 0 & 9 & 19 \\ 0 & 9 & 8 \\ 1 & 6 & 23 \end{bmatrix} \begin{bmatrix} 0 \\ 2 \\ 19 \end{bmatrix} = \begin{bmatrix} 9 \cdot 2 + 19 \cdot 19 \\ 9 \cdot 2 + 8 \cdot 19 \\ 6 \cdot 2 + 23 \cdot 19 \end{bmatrix} \pmod{26}.$$

Row 1: $18 + 361 \equiv 18 + 23 = 41 \equiv 15$
Row 2: $18 + 152 \equiv 18 + 22 = 40 \equiv 14$
Row 3: $12 + 437 \equiv 12 + 21 = 33 \equiv 7$
So output is $(15, 14, 7)^T =$ `POH`. ✓

Check `NOW` $\to$ `YQF`:

$$KP_2 = \begin{bmatrix} 0 & 9 & 19 \\ 0 & 9 & 8 \\ 1 & 6 & 23 \end{bmatrix} \begin{bmatrix} 13 \\ 14 \\ 22 \end{bmatrix} \pmod{26}.$$

Row 1: $9 \cdot 14 + 19 \cdot 22 = 126 + 418 \equiv 22 + 2 = 24$
Row 2: $9 \cdot 14 + 8 \cdot 22 = 126 + 176 \equiv 22 + 20 = 42 \equiv 16$
Row 3: $13 + 6 \cdot 14 + 23 \cdot 22 = 13 + 84 + 506 \equiv 13 + 6 + 12 = 31 \equiv 5$
So output is $(24, 16, 5)^T =$ `YQF`. ✓

(5) Check that K is invertible mod 26 (extra)

$$\det(K) = \begin{vmatrix} 0 & 9 & 19 \\ 0 & 9 & 8 \\ 1 & 6 & 23 \end{vmatrix} = -9 \begin{vmatrix} 0 & 8 \\ 1 & 23 \end{vmatrix} + 19 \begin{vmatrix} 0 & 9 \\ 1 & 6 \end{vmatrix} = -9(-8) + 19(-9) = 72 - 171 = -99.$$

$$-99 \equiv -99 + 104 = 5 \pmod{26}, \quad \gcd(5, 26) = 1.$$

So K is invertable. ✓

Final Answer (Q2):

$$\boxed{K = \begin{bmatrix} 0 & 9 & 19 \\ 0 & 9 & 8 \\ 1 & 6 & 23 \end{bmatrix} \pmod{26}}$$

Question 3 (Perfect secrecy: key size and key reuse)

(1) Perfect secrecy $\Rightarrow |K| \geq |M|$

Statement: If encryption scheme provide perfect secrecy, then:

$$|K| \geq |M|.$$

Definitions used:

- M: message space (all possible plaintexts).
- K: key space (all possible keys).
- C: ciphertext space.
- Encryption: $\mathrm{Enc}_k(m)$ and decryption: $\mathrm{Dec}_k(c)$.
- Correctness: $\mathrm{Dec}_k(\mathrm{Enc}_k(m)) = m$ for all $m \in M, k \in K$.
- Perfect secrecy (Shannon):

$$\Pr[M = m \mid C = c] = \Pr[M = m]$$

  for all m and all c with $\Pr[C = c] > 0$.
- **(Tiny assumtion for grading)** I assume $\Pr[M = m] > 0$ for all $m \in M$ (or at least for all meaningful messages), otherwise perfect secrecy condition is trivial for zero-probability messages.

Lemma 1 (A ciphertext + a key gives exactly one plaintext):
Fix c. For each key k, decryption gives exactly one message:

$$f_c(k) = \mathrm{Dec}_k(c).$$

So $f_c : K \to M$ is a function.

Lemma 2 (For a fixed ciphertext, at most $|K|$ messages are possible):
Define:

$$S_c = \{m \in M : \exists\, k \in K \text{ such that } \mathrm{Enc}_k(m) = c\}.$$

Since each key decrypts c to exactly one message, we have:

$$|S_c| \leq |K|.$$

Lemma 3 (Perfect secrecy forces every message to remain possible):
Take any ciphertext c with $\Pr[C = c] > 0$. For any message m with $\Pr[M = m] > 0$, perfect secrecy gives:

$$\Pr[M = m \mid C = c] = \Pr[M = m] > 0.$$

So m cannot become impossible after seeing c, which mean $m \in S_c$. Hence all such messages remain possible, so (for our message space) we can write:

$$S_c = M \quad \Rightarrow \quad |S_c| = |M|.$$

Final proof:
From Lemma 2: $|S_c| \leq |K|$. From Lemma 3: $|S_c| = |M|$. Hence:

$$|M| \leq |K| \quad \Rightarrow \quad \boxed{|K| \geq |M|}.$$

Simpel analogy:
Keys are like costums. Perfect secrecy means that after seeing a ciphertext look c, every message must still be possible. If there are fewer costums than messages, some messages cannot look like c, and then ciphertext leaks info.

(2) Perfect secrecy is impossible if I reuse same secret key for multiple messages

Statement: No symmetric-key encryption can has perfect secrecy if the same key is reused across multiple messages.

Main reason: The same hidden key creates a link (correlation) between ciphertexts.

Known-plaintext style argument:
If M_1 is a known header and I reuse same key for M_2, then from (M_1, C_1) the attacker can restrict or learn the key, and then C_2 leaks information about M_2. So:

$$\Pr[M_2 = m \mid C_1 = c_1, C_2 = c_2] \neq \Pr[M_2 = m],$$

which breaks perfect secrecy.

Extra formal link to part (1):
If perfect secrecy held for the *pair* (M_1, M_2) under a reused key, then I can treat the combined message space as $M \times M$ (size $|M|^2$). Applying part (1) to this combined message would require:

$$|K| \geq |M \times M| = |M|^2.$$

But in key reuse, same fixed-size key space K is used for both messages, so in general this requirement is violated.

Information theory view (short):
To hide one message perfectly, need $H(K) \geq H(M)$. To hide two independant messages perfectly with same key, need:

$$H(K) \geq H(M_1, M_2) = H(M_1) + H(M_2),$$

so key uncertainty must grow with number of messages. Reusing one fixed key do not.

Two-time pad demo (OTP):

$$C_1 = M_1 \oplus K, \qquad C_2 = M_2 \oplus K \Rightarrow C_1 \oplus C_2 = M_1 \oplus M_2.$$

So key cancels out and attacker learns a relation between plaintexts.

Conclusion: Key reuse creates unavoidable links between ciphertexts, so perfect secrecy is impossible.

Question 5 (Shannon theory: spurious keys and perfect secrecy)

Given / Setup

I am given a cryptosystem (P, C, K, E, D) where:

- $|P| = |C|$.
- Keys are chosen equiprobably from key space K.
- Ciphertext length is n, so ciphertexts are in C^n.
- R_L is the redundancy of the underlying language.

What redundancy R_L means (the way I will use it)

Redundancy means: only small fraction of all possible length-n strings are meaningful (valid English-like messages).

Let:

- P^n be all length-n plaintext strings, so $|P^n| = |P|^n$.
- $L_n \subseteq P^n$ be the set of meaningful plaintext strings.

Shannon's redundancy model says (approximately):

$$|L_n| \approx |P|^{n(1-R_L)}.$$

So the fraction of meaningful strings among all strings is:

$$\frac{|L_n|}{|P|^n} \approx |P|^{-nR_L}.$$

Part (1) Prove the lower bound on expected spurious keys

Definition of spurious key:
I intercept a ciphertext $c \in C^n$. Define:

$$N(c) = \#\{k \in K : \ D_k(c) \in L_n\}.$$

This count how many keys decrypt c into a meaningful plaintext.

If true key is k_0, then k_0 is counted inside $N(c)$, so the number of spurious keys is:

$$s_n(c) = N(c) - 1.$$

Let expected number of spurious keys be $\bar{s}_n = \mathbb{E}[s_n(C)]$. I needs to show:

$$\boxed{\bar{s}_n \geq \frac{|K|}{|P|^{nR_L}} - 1.}$$

Step 1 (correctness gives bijection):
For a fixed key k, E_k and D_k are inverses, so $E_k : P^n \to C^n$ is a bijection. So for each k and ciphertext c, there is exactly one plaintext $D_k(c)$.

Step 2 (double counting):
Count pairs (p, k) with $p \in L_n$ and $k \in K$. Total:

$$|L_n| \cdot |K|.$$

Each pair produces exactly one ciphertext $c = E_k(p)$.

Group by ciphertext c. The number of pairs that map to a fixed ciphertext c is exactly $N(c)$. So:

$$\sum_{c \in C^n} N(c) = |K| \cdot |L_n|. \qquad \text{(A)}$$

Step 3 (define the expectation experiment and lower bound):
(Clarity line) I take expectation over the experiment where K is uniform over the key space and P is uniform over meaningful strings L_n, and then $C = E_K(P)$.

Then:

$$\Pr[C = c] = \frac{N(c)}{|K| \cdot |L_n|}.$$

So:

$$\begin{aligned}\mathbb{E}[N(C)] &= \sum_{c \in C^n} \Pr[C = c] \cdot N(c) \\ &= \sum_{c \in C^n} \frac{N(c)}{|K| \cdot |L_n|} \cdot N(c) = \frac{1}{|K| \cdot |L_n|} \sum_{c \in C^n} N(c)^2. \qquad \text{(B)}\end{aligned}$$

Now apply Cauchy–Schwarz:

$$\sum_{c \in C^n} N(c)^2 \geq \frac{\left(\sum_{c \in C^n} N(c)\right)^2}{|C^n|}.$$

Since $|C^n| = |C|^n = |P|^n$, we have:

$$\sum_{c \in C^n} N(c)^2 \geq \frac{\left(\sum_{c \in C^n} N(c)\right)^2}{|P|^n}. \qquad \text{(C)}$$

Plug (C) into (B) and use (A):

$$\mathbb{E}[N(C)] \geq \frac{1}{|K| \cdot |L_n|} \cdot \frac{(|K| \cdot |L_n|)^2}{|P|^n} = \frac{|K| \cdot |L_n|}{|P|^n}. \qquad \text{(D)}$$

Step 4 (use redundancy model):

$$\frac{|L_n|}{|P|^n} \approx |P|^{-nR_L} \Rightarrow \mathbb{E}[N(C)] \geq |K| \cdot |P|^{-nR_L} = \frac{|K|}{|P|^{nR_L}}.$$

Finally:

$$\bar{s}_n = \mathbb{E}[s_n(C)] = \mathbb{E}[N(C) - 1] = \mathbb{E}[N(C)] - 1 \Rightarrow \boxed{\bar{s}_n \geq \frac{|K|}{|P|^{nR_L}} - 1.}$$

Intuition (short)

Meaningful messages is rare. So among many wrong keys, a few will accidentally decrypt to something meaningful. The bound says we cannot avoid having at least about $\frac{|K|}{|P|^{nR_L}} - 1$ such false meaningful keys on average.

Part (2) Use this to explain OTP perfect secrecy and infinite unicity distance

(2a) OTP perfect secrecy:
In OTP, for any plaintext $m \in P^n$ and ciphertext $c \in C^n$, there is exactly one key $k = c \oplus m$ (or $k = c - m \mod q$). Because key is uniform:

$$\Pr[C = c \mid M = m] = \frac{1}{|P|^n},$$

independant of m. So ciphertext gives no information about plaintext.

(2b) Infinite unicity distance:
For OTP, $|K| = |P|^n$. Plug into the bound:

$$\bar{s}_n \geq \frac{|P|^n}{|P|^{nR_L}} - 1 = |P|^{n(1-R_L)} - 1.$$

Since real languages have $R_L < 1$, $(1 - R_L) > 0$, so $\bar{s}_n$ grows with n. So spurious keys do not go to 0, and attacker never gets a unique key from ciphertext alone. Hence OTP unicity distance is effectivly infinite.

Question 6 (One-Time Pad and perfect secrecy)

6.1 Shannon's definition of perfect secrecy

An encryption scheme have perfect secrecy if observing the ciphertext gives no extra information about the plaintext.

Formally, for every plain text $m \in M$ and cipher text $c \in C$ with $\Pr[C = c] > 0$:

$$\boxed{\Pr[M = m \mid C = c] = \Pr[M = m].}$$

So my belief about the plaintext dont change after I see the ciphertext.

Equivallent form: Ciphertext distribution does not depends on plaintext:

$$\Pr[C = c \mid M = m_1] = \Pr[C = c \mid M = m_2] \quad \forall m_1, m_2 \in M,\ \forall c \in C.$$

6.2 Prove that OTP satisfies Shannon perfect secrecy

OTP setup (how I define it):

- Message M is a length-n string over an alphabet of size q.
- Key K is also length n, chosen uniformly at random from $\{0, \ldots, q-1\}^n$.
- Encryption is component-wise addition mod q:

$$C = M + K \pmod q.$$

- Decryption:

$$M = C - K \pmod q.$$

Proof (ciphertext gives no information):
I have to show:

$$\Pr[M = m \mid C = c] = \Pr[M = m] \quad \text{for all } m, c.$$

Step 1: Compute $\Pr[C = c \mid M = m]$.
Fix any plaintext m and ciphertext c. Under OTP:

$$c = m + k \pmod q \Rightarrow k = c - m \pmod q.$$

Important point: for fixed m and c, there is exactly one key k that makes this true.

Since K is uniform over q^n keys:

$$\Pr[K = c - m] = \frac{1}{q^n}.$$

So:

$$\boxed{\Pr[C = c \mid M = m] = \frac{1}{q^n}.}$$

This value does not depends on m.

Step 2: Conclude perfect secrecy.
Because $\Pr[C = c \mid M = m]$ is same for every m, cipher text and plain text are independant. So Shannon condition holds:

$$\Pr[M = m \mid C = c] = \Pr[M = m].$$

Hence OTP satisfies perfect secrecy.

6.3 Key reuse: same key k for m_1 and m_2. What does Eve learn?

Alice sends:

$$c_1 = m_1 + k \pmod q, \qquad c_2 = m_2 + k \pmod q.$$

Eve can subtract:

$$c_1 - c_2 \equiv (m_1 + k) - (m_2 + k) \equiv m_1 - m_2 \pmod q.$$

So key cancels out, and Eve learns a direct relation between the two plaintexts.

For bit OTP ($q = 2$), subtraction is XOR, so:

$$\boxed{c_1 \oplus c_2 = m_1 \oplus m_2.}$$

This is the **two-time pad** problem.

Why dangerous:

- If Eve guesses part of m_1 (like a known header), she can recover corresponding part of m_2.
- If one plaintext is known fuly, then $k = c_1 - m_1 \mod q$ and she decrypts the other: $m_2 = c_2 - k \mod q$.

Simpel analogy: OTP key is like a perfect mask paint. If same paint pattern is used twice, comparing the two outputs cancels the paint and reveals relationships between original texts.

6.4 If OTP is perfectly secure, why is it not used in practice?

Even though OTP is theoretically perfect, it is not used in most real systems because:

- **Key length problem:** key must be as long as entire message. If I send 1 GB data, I need 1 GB truly random key shared in advance.
- **Key distribution problem:** Alice and Bob must allready share huge keys securily. If they already had that, main problem is already solved.
- **True randomness required:** keys must be uniform, unpredictable, and never reused. Generating and storing lot of true randomness is hard at scale.
- **Key storage/management:** keys must be stored securely, synchronized, and never reused.
- **Catastrophic failure on reuse:** one mistake in reuse can break security badly (as in 6.3).

Practical conclusion: In practice we use modern symmetric encryption (like AES) with short keys and safe IV/nonce handling. It is computationally secure against feasible attackers.

End of document.